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1.

Kotlin

```
data class Mahasiswa(val nama: String, val email: String?)

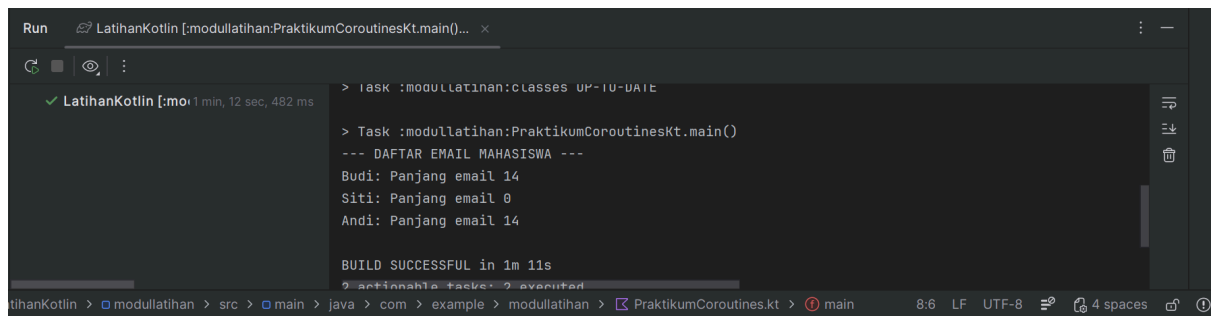
fun main() {
    val daftarMhs = listOf(
        Mahasiswa("Budi", "budi@gmail.com"),
        Mahasiswa("Siti", null), // Email kosong
        Mahasiswa("Andi", "andi@yahoo.com")
    )

    println("--- DAFTAR EMAIL MAHASISWA ---")

    for (mhs in daftarMhs) {
        // TUGAS 1: Ambil panjang karakter email.
        // Menggunakan Safe Call (?.) dan Elvis Operator (?:)
        val panjangEmail = mhs.email?.length ?: 0

        println("${mhs.nama}: Panjang email $panjangEmail")
    }
}
```

output:



```
Run LatihanKotlin [:modullatihan:PraktikumCoroutinesKt.main()... x
✓ LatihanKotlin [:mod 1 min, 12 sec, 482 ms
> Task :modullatihan:classes UP-TO-DATE
> Task :modullatihan:PraktikumCoroutinesKt.main()
--- DAFTAR EMAIL MAHASISWA ---
Budi: Panjang email 14
Siti: Panjang email 0
Andi: Panjang email 14
BUILD SUCCESSFUL in 1m 11s
2 actionable tasks: 2 executed
LatihanKotlin > modullatihan > src > main > java > com > example > modullatihan > PraktikumCoroutines.kt > main 8:6 LF UTF-8 4 spaces
```

2.

Kotlin

```
import kotlinx.coroutines.*
import kotlin.system.measureTimeMillis

// Simulasi API call (JANGAN DIUBAH)
suspend fun getProfil(): String {
    delay(1000) // Pura-pura loading 1 detik
}
```

```

        return "Profil: Budi Santoso"
    }

suspend fun getTransaksi(): String {
    delay(1000) // Pura-pura loading 1 detik
    return "Transaksi: [Beli Laptop, Beli Mouse]"
}

fun main() = runBlocking {
    println("--- MULAI DOWNLOAD DATA ---")

    val waktu = measureTimeMillis {
        // TUGAS 2: Ubah kode di bawah ini agar berjalan PARALEL (Bersamaan)
        // Menggunakan async dan await

        val deferredProfil = async { getProfil() }
        val deferredTransaksi = async { getTransaksi() }

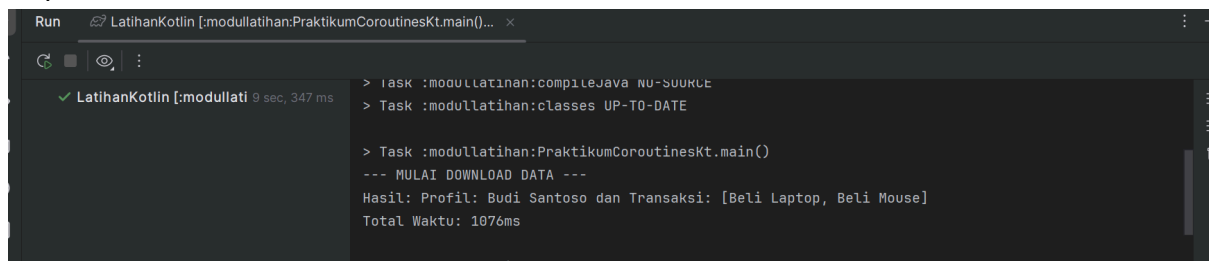
        val profil = deferredProfil.await()
        val transaksi = deferredTransaksi.await()

        println("Hasil: $profil dan $transaksi")
    }

    println("Total Waktu: ${waktu}ms")
}

```

output :



```

Run LatihanKotlin [modullatihan:PraktikumCoroutinesKt.main()]...
✓ LatihanKotlin [modullati 9 sec, 347 ms]
> Task :modullatihan:compileJava NO-SOURCE
> Task :modullatihan:classes UP-TO-DATE
> Task :modullatihan:PraktikumCoroutinesKt.main()
--- MULAI DOWNLOAD DATA ---
Hasil: Profil: Budi Santoso dan Transaksi: [Beli Laptop, Beli Mouse]
Total Waktu: 1076ms
BUILD SUCCESSFUL in 9s

```

3.

Kotlin

```

import kotlinx.coroutines.*
import kotlinx.coroutines.flow.*

// Simulasi Sensor (Producer)
fun sensorSuhu(): Flow<Int> = flow {
    val dataSuhu = listOf(28, 30, 36, 29, 40, 32, 45)
    for (suhu in dataSuhu) {

```

```

        delay(500) // Sensor update tiap 0.5 detik
        emit(suhu) // Kirim data
    }
}

fun main() = runBlocking {
    println("--- MONITOR SUHU AKTIF ---")

    sensorSuhu()
        // TUGAS 3a: Filter data. Hanya loloskan suhu > 35
        .filter { suhu -> suhu > 35 }
        // TUGAS 3b: Ubah format data jadi String "PERINGATAN! Suhu: X
        derajat"
        .map { suhuPanas -> "PERINGATAN! Suhu: $suhuPanas derajat" }
        .collect { pesan ->
            println(pesan)
        }
}

```

output :

```

Run LatihanKotlin [:modullatihan:PraktikumFlowKt.main()] x
> Task :modullatihan:classes UP-TO-DATE
> Task :modullatihan:PraktikumFlowKt.main()
--- MONITOR SUHU AKTIF ---
PERINGATAN! Suhu: 36 derajat
PERINGATAN! Suhu: 40 derajat
PERINGATAN! Suhu: 45 derajat
BUILD SUCCESSFUL in 9s
9 actionable tasks: 2 executed

```